

Interactive Graphics

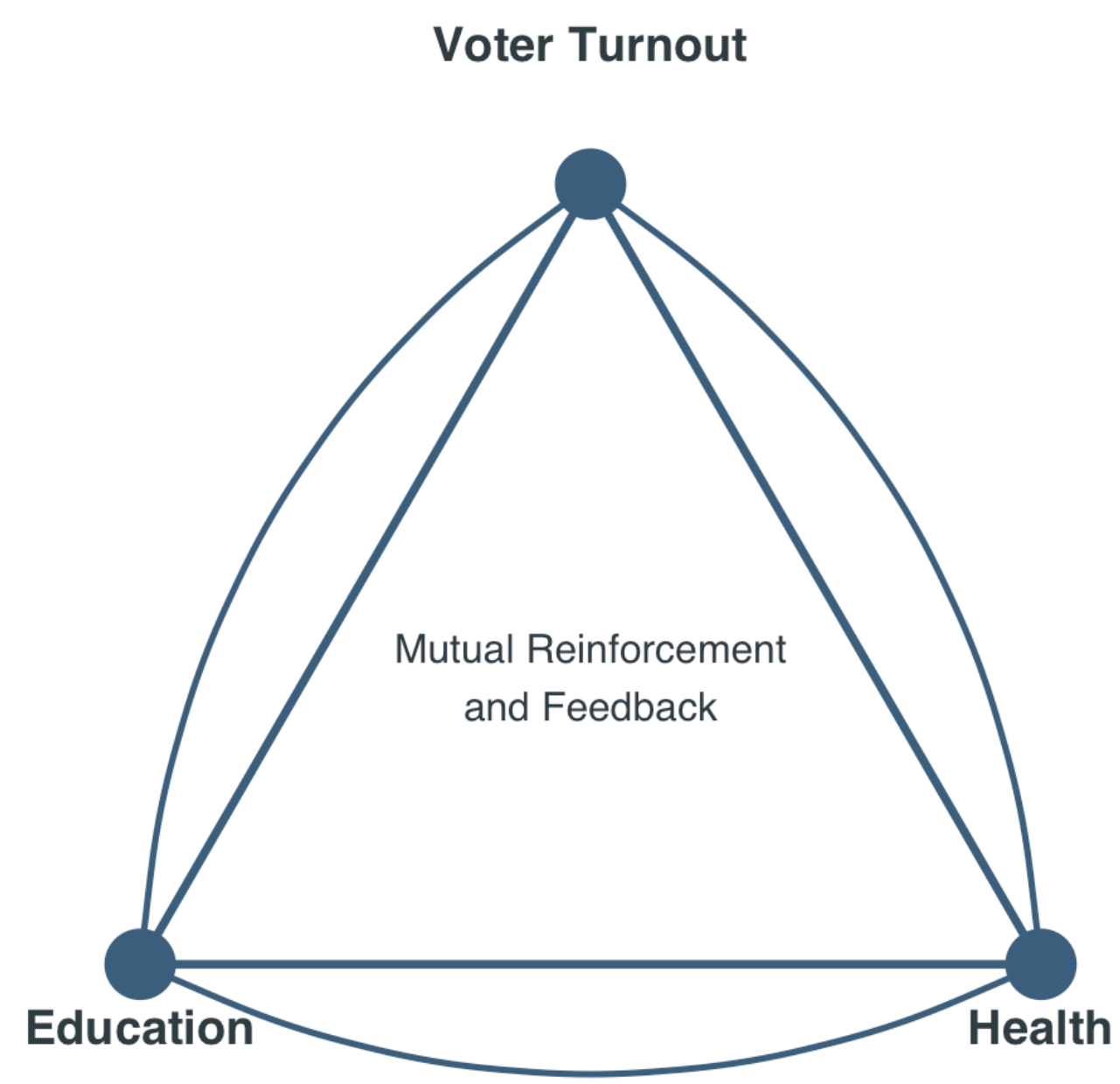
<https://john-glendenning.shinyapps.io/FinalReportGraphics/>

Mapping Civic Inequality in Texas:

Gardens and Deserts of Democratic Engagement

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The Civic Triangle: Education, Health, and Voter Turnout



Expected Relationship	
Education → Voting	Education increases political involvement, civic knowledge, and voter participation.
Health → Voting	Healthier individuals are more likely to vote and engage. Poor health reduces civic participation.
Education → Health	Education predicts life expectancy, preventive care use, and health literacy.
Voting → Health Policy	Communities with higher turnout have more influence on health and education policy outcomes.
Health → Education	Healthy people perform better academically; adult wellness supports lifelong learning.
Voting → Education	Participatory democracies sustain educational investment; educated voters support public health policy.

The Civic Triangle

Data and Methodology

Texas State Data

(2 variables)

- Voter Turnout
- Registered Voters

American Community Survey

(23 variables)

- Level of Education

CDC Places

(8 variables)

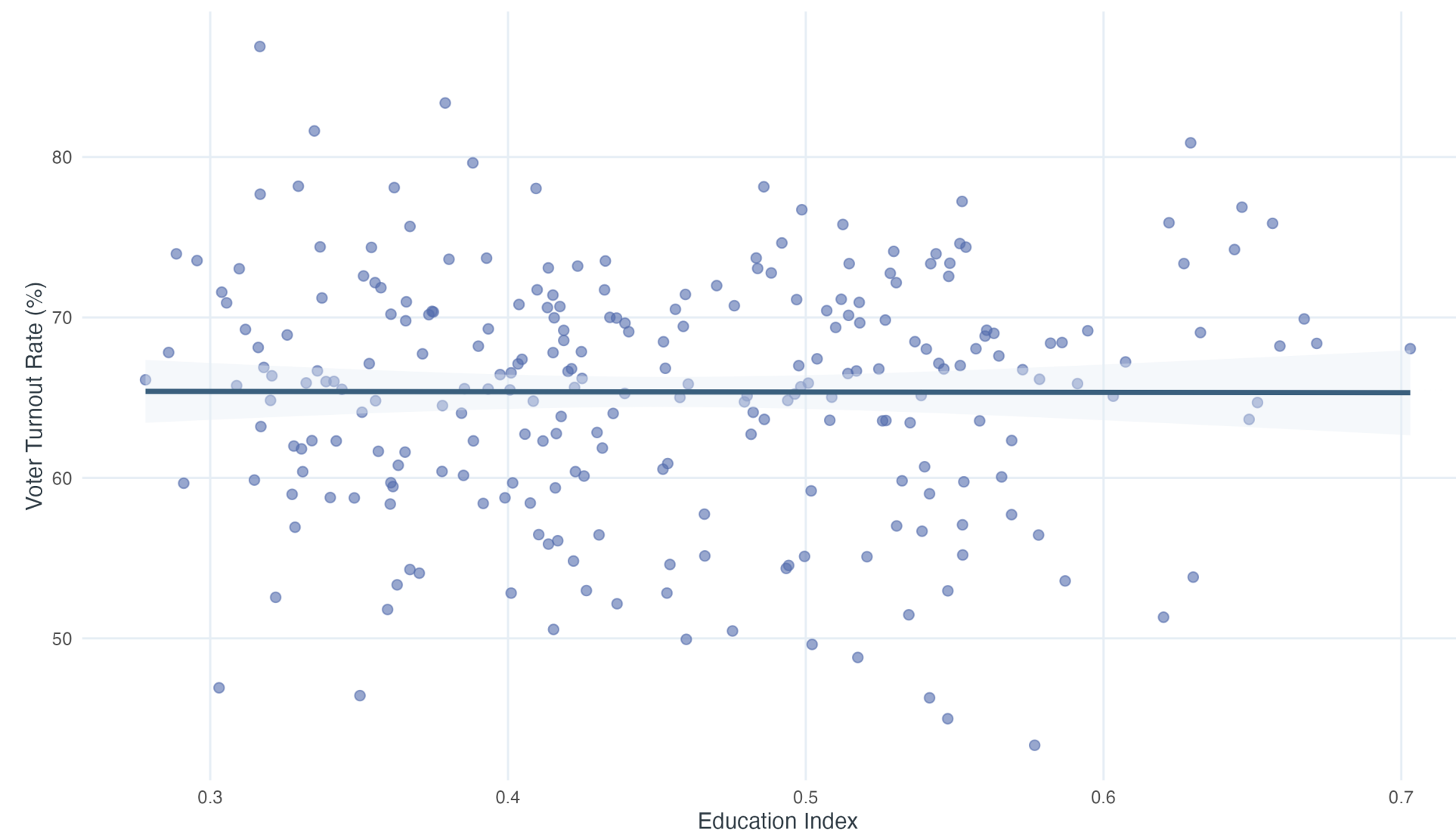
- General & Physical Health
- Mental Health
- Health Insurance and Checkups
- Obesity



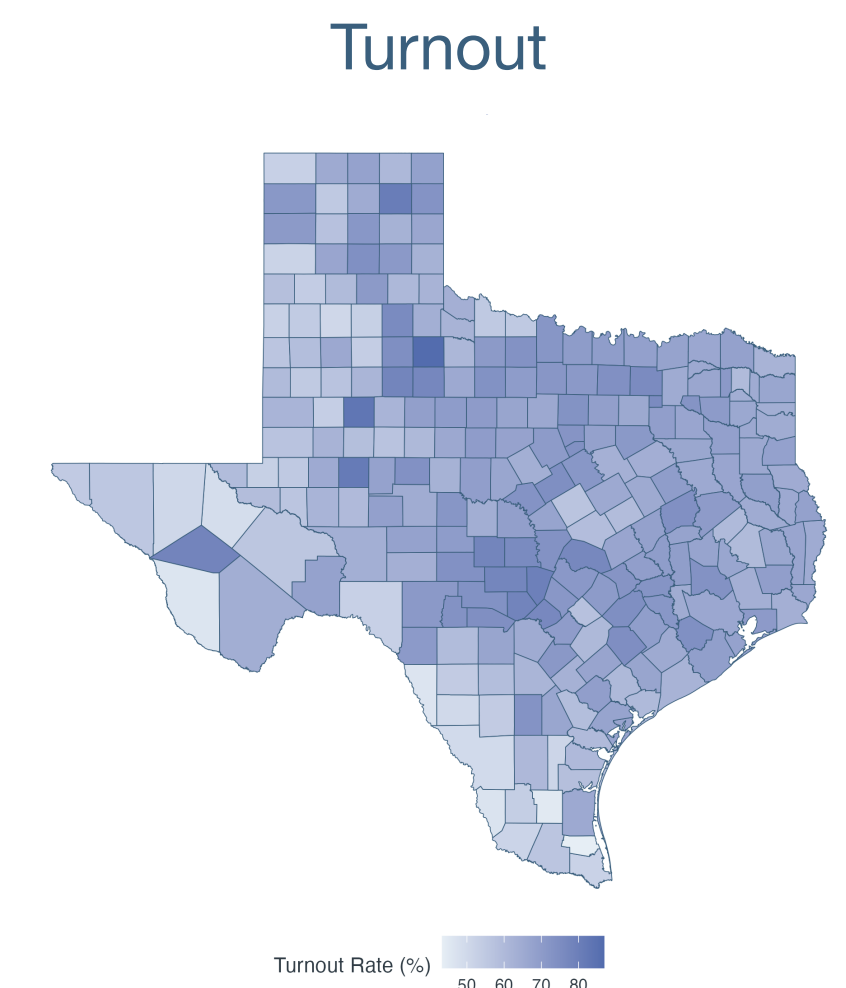
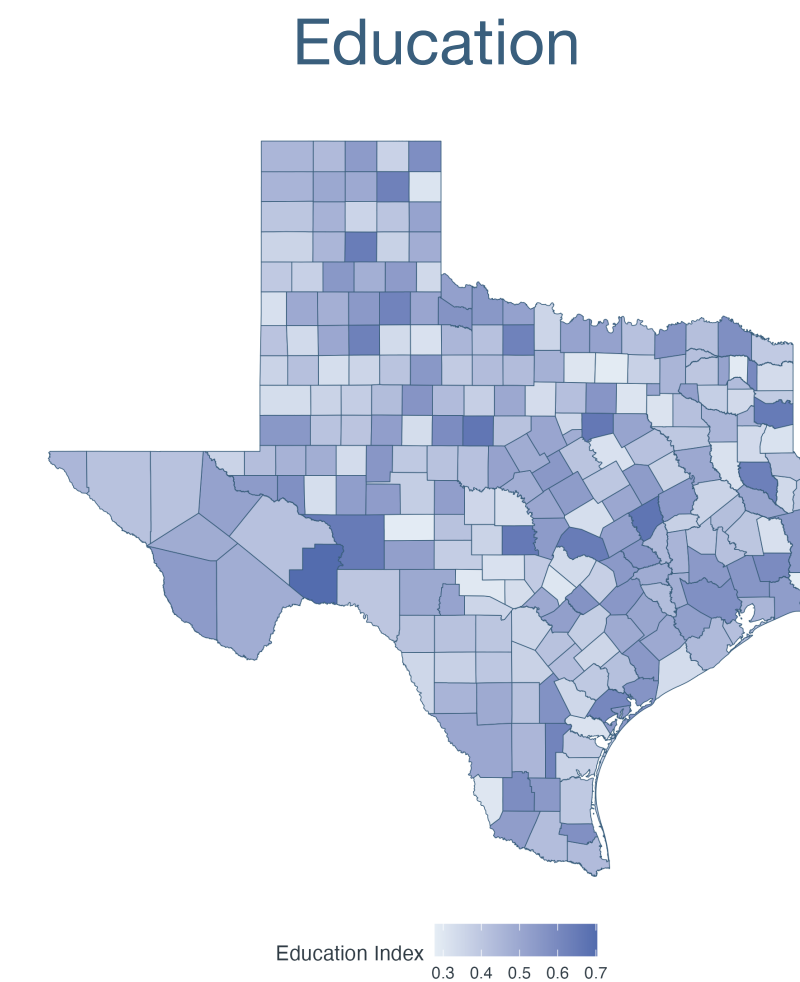
Civic Vitality Index

$$\text{CivicIndex}_i = \frac{\frac{\text{TurnoutRate}_i}{100} + \text{EduIndex}_i + (1 - \text{HealthIndex}_i)}{3}$$

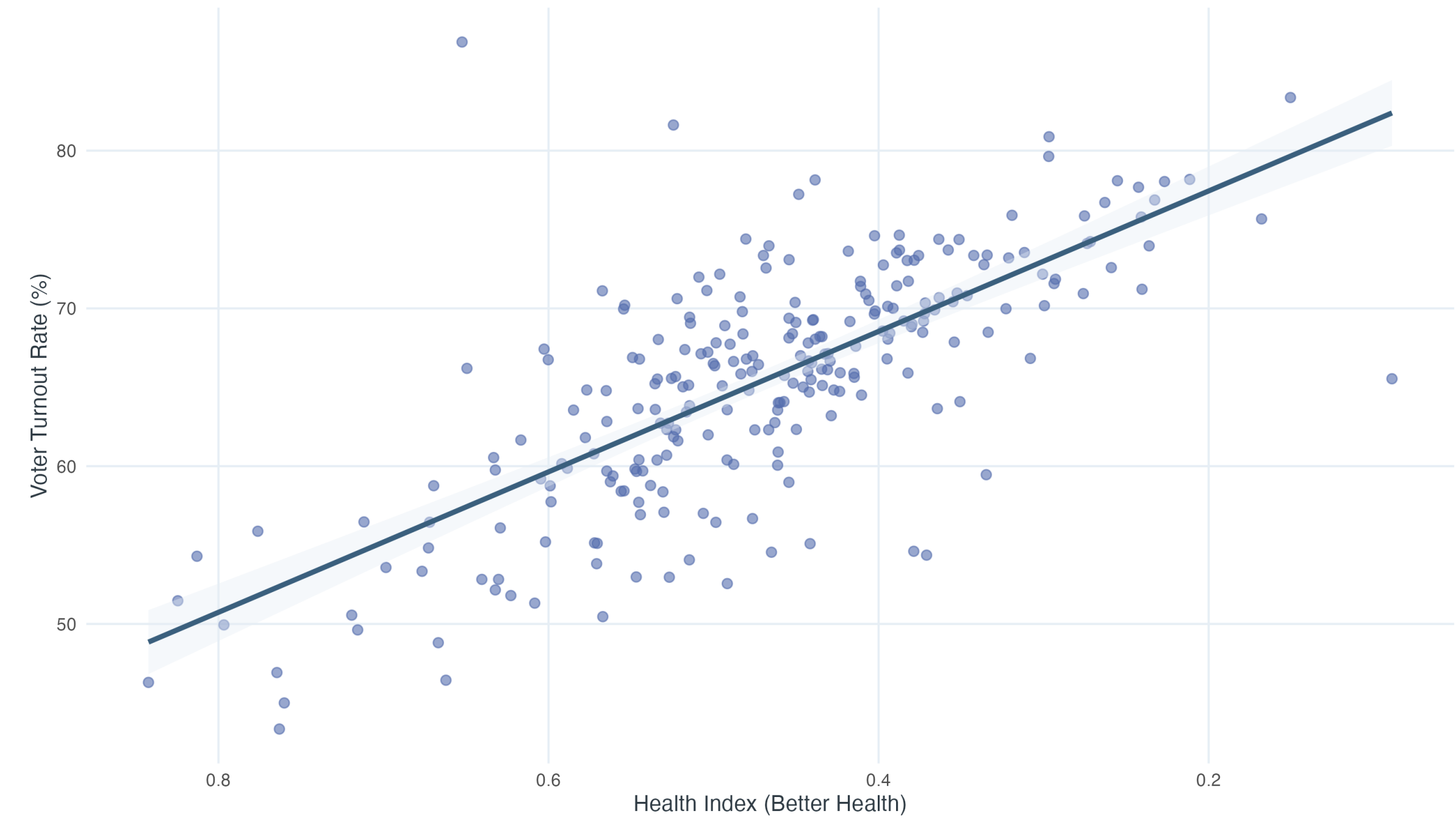
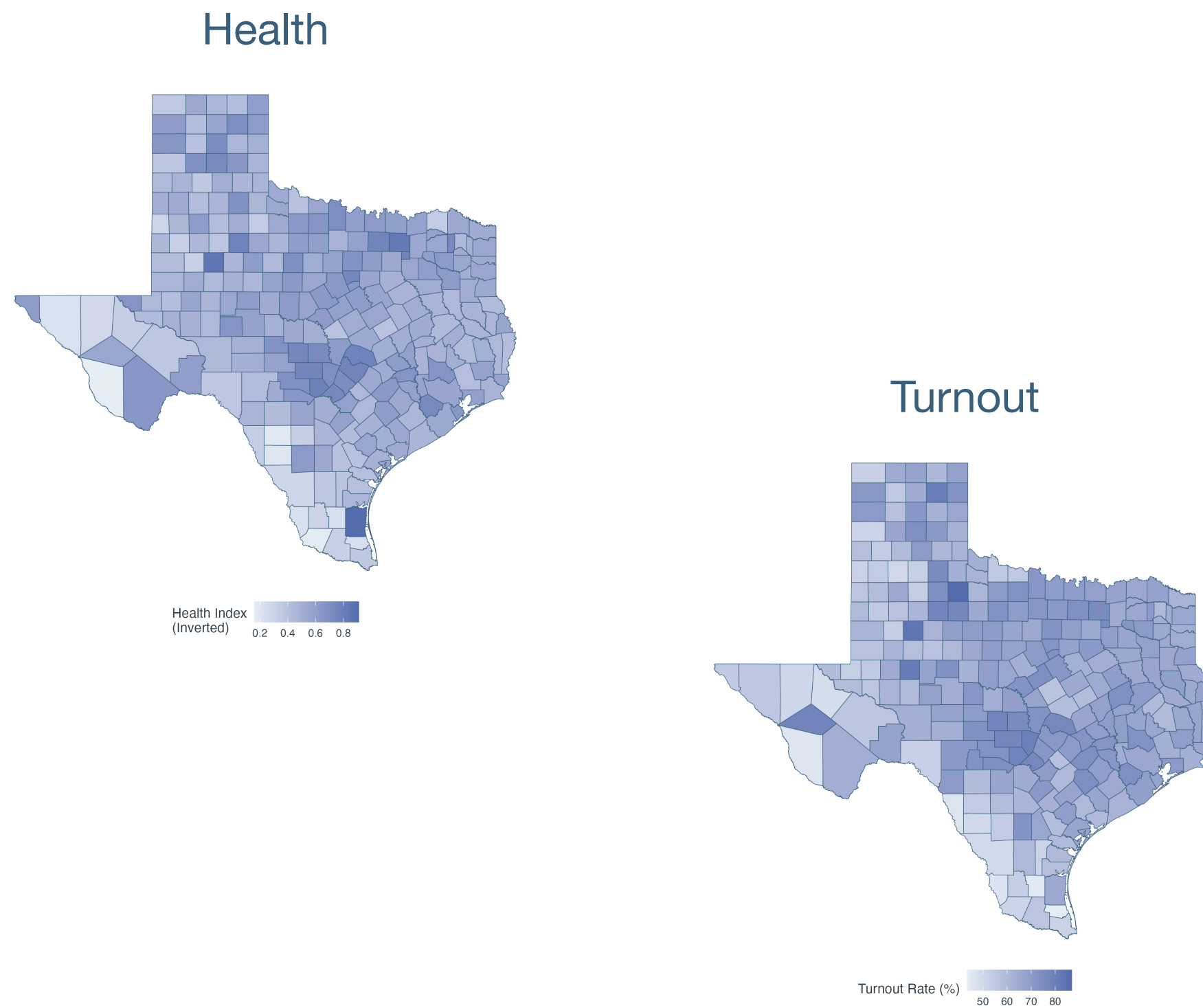
Education Surprise



- Weak Correlation between Turnout and Education
- Spatially Disjointed
- Index and Turnout maps would be similar if linked

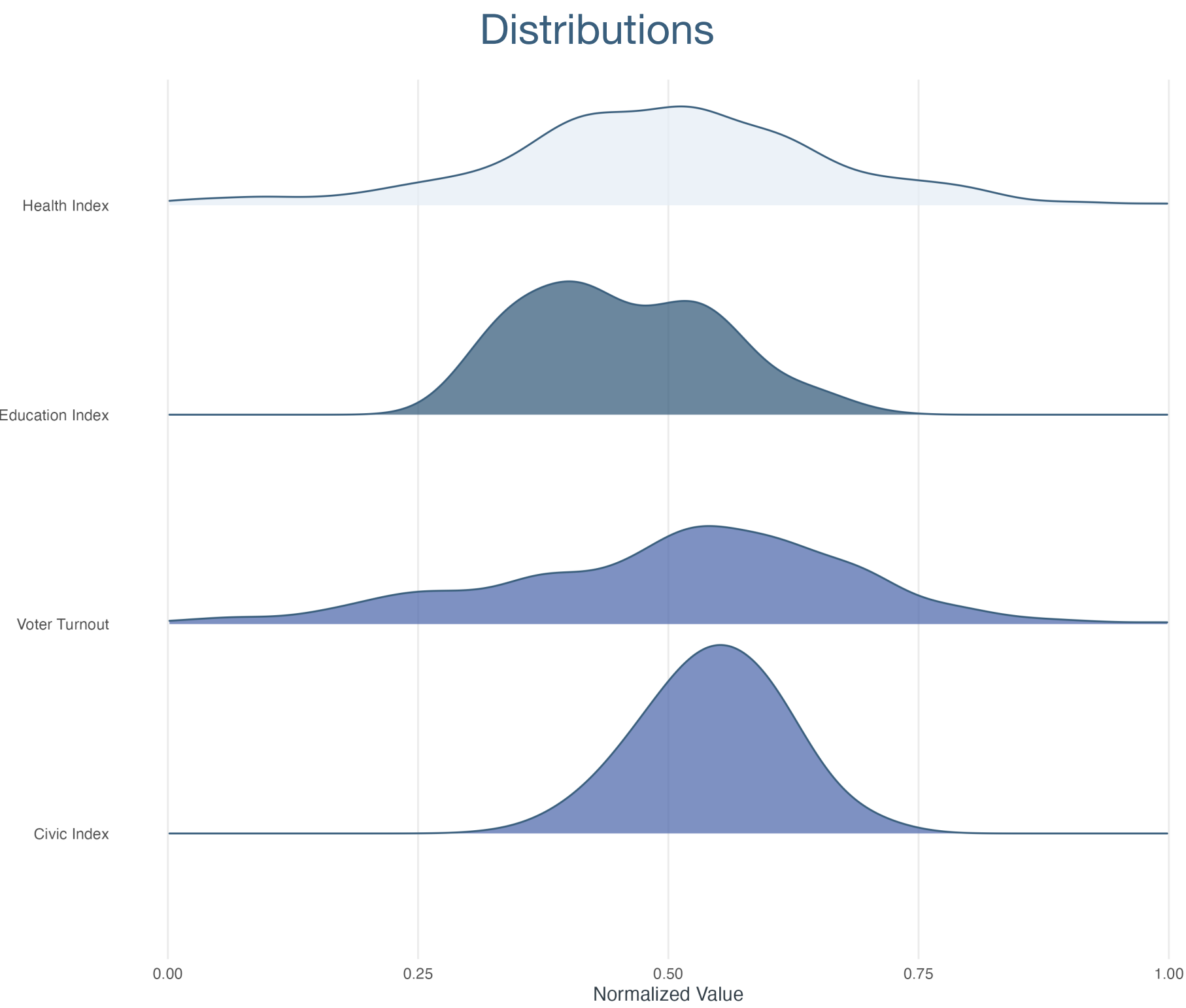
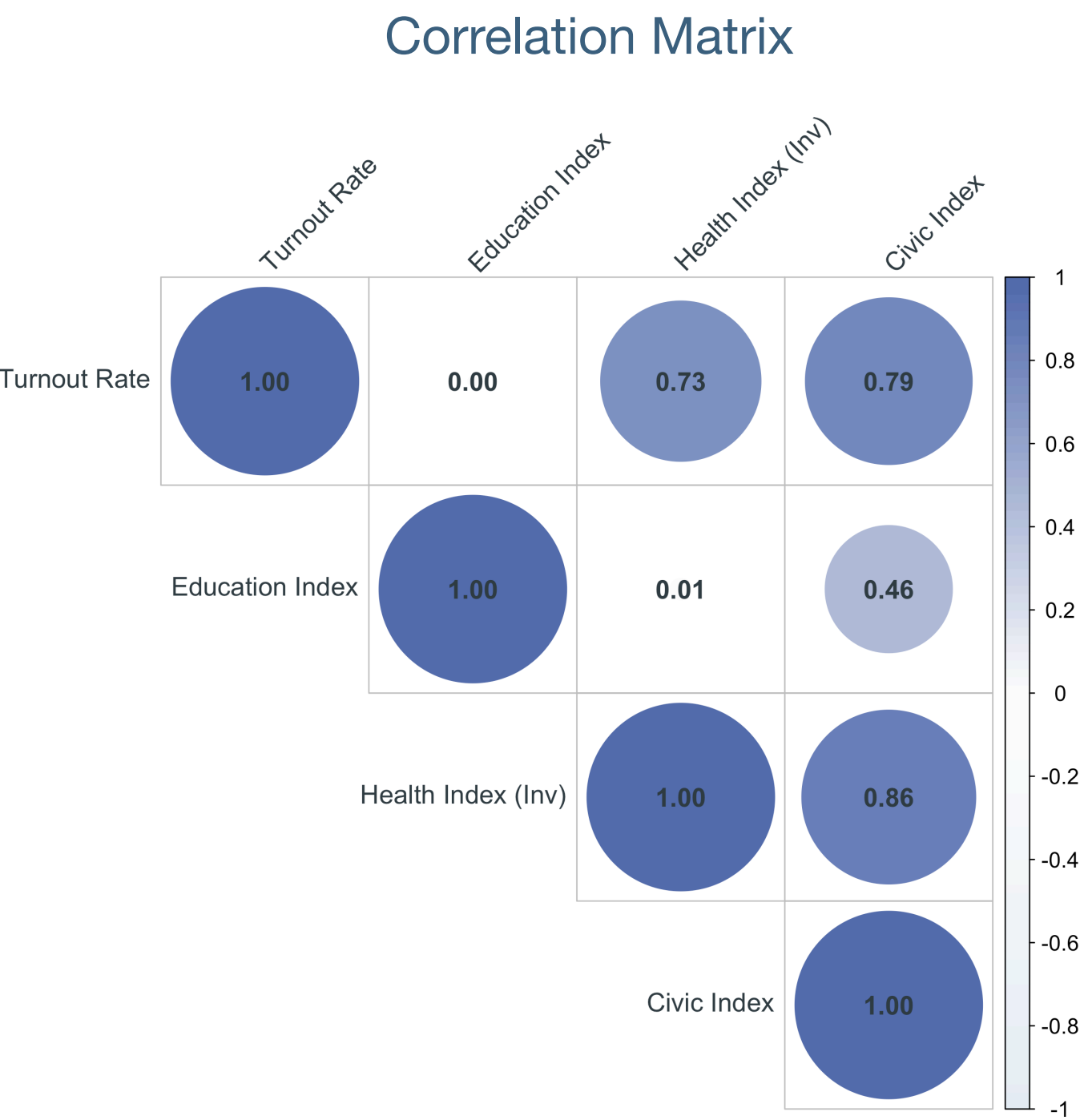


Health is Key

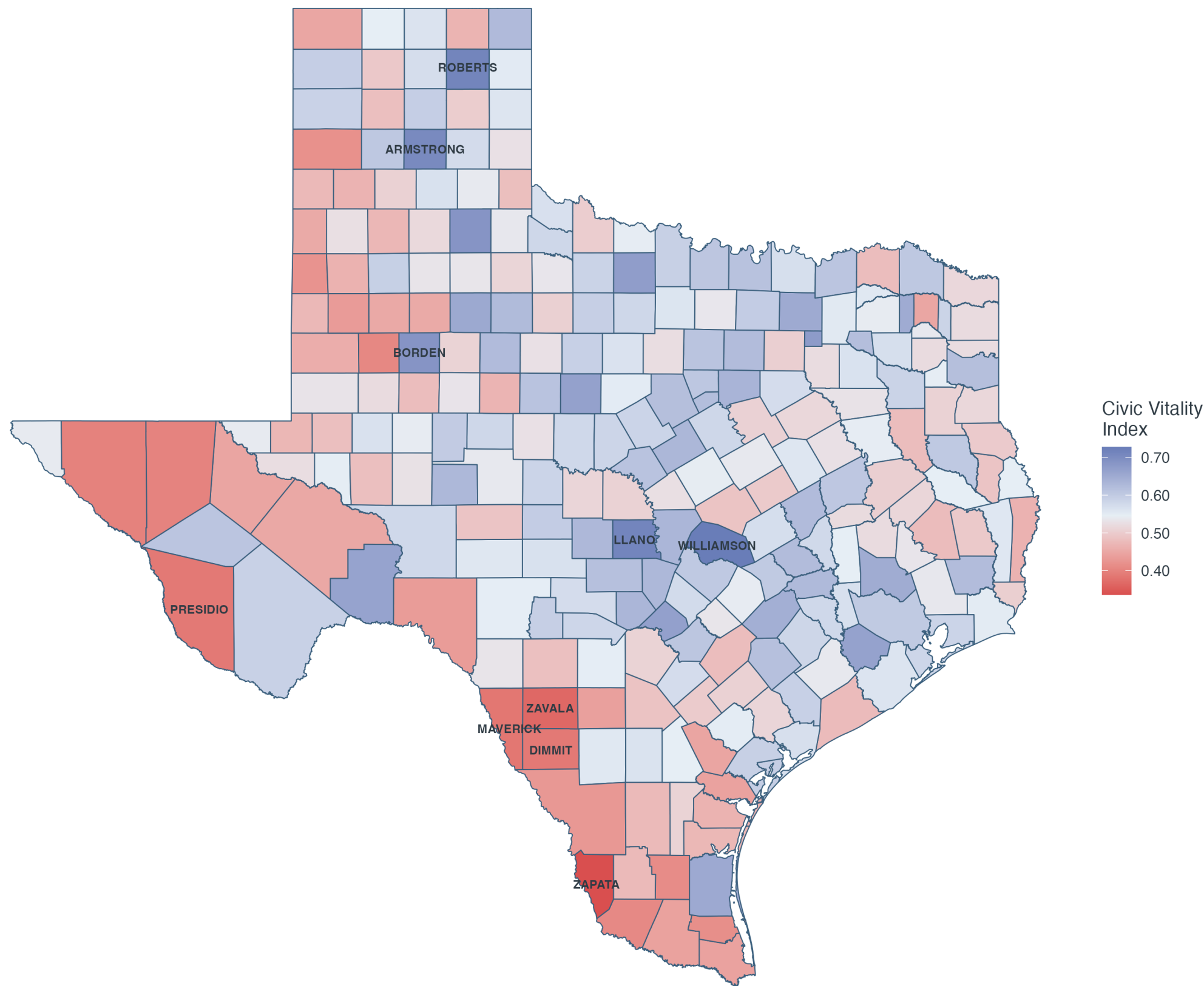


- Health is a stronger predictor
- Spatially Similar

Interrelationships



Geographic Patterns



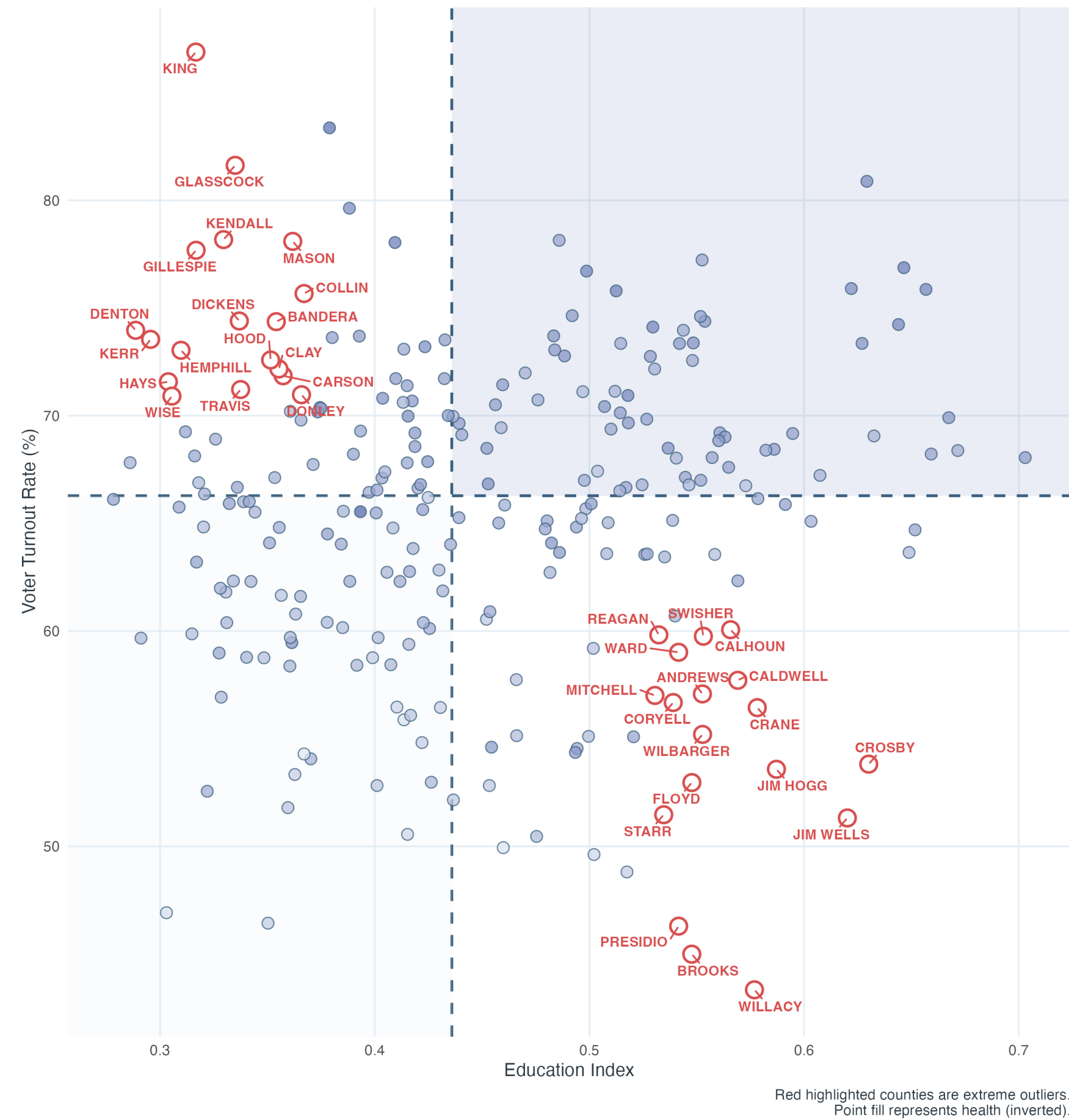
Higher values indicate stronger civic vitality. Top and bottom 5 counties labeled.

Civic Vitality Index

Quadrant Analysis

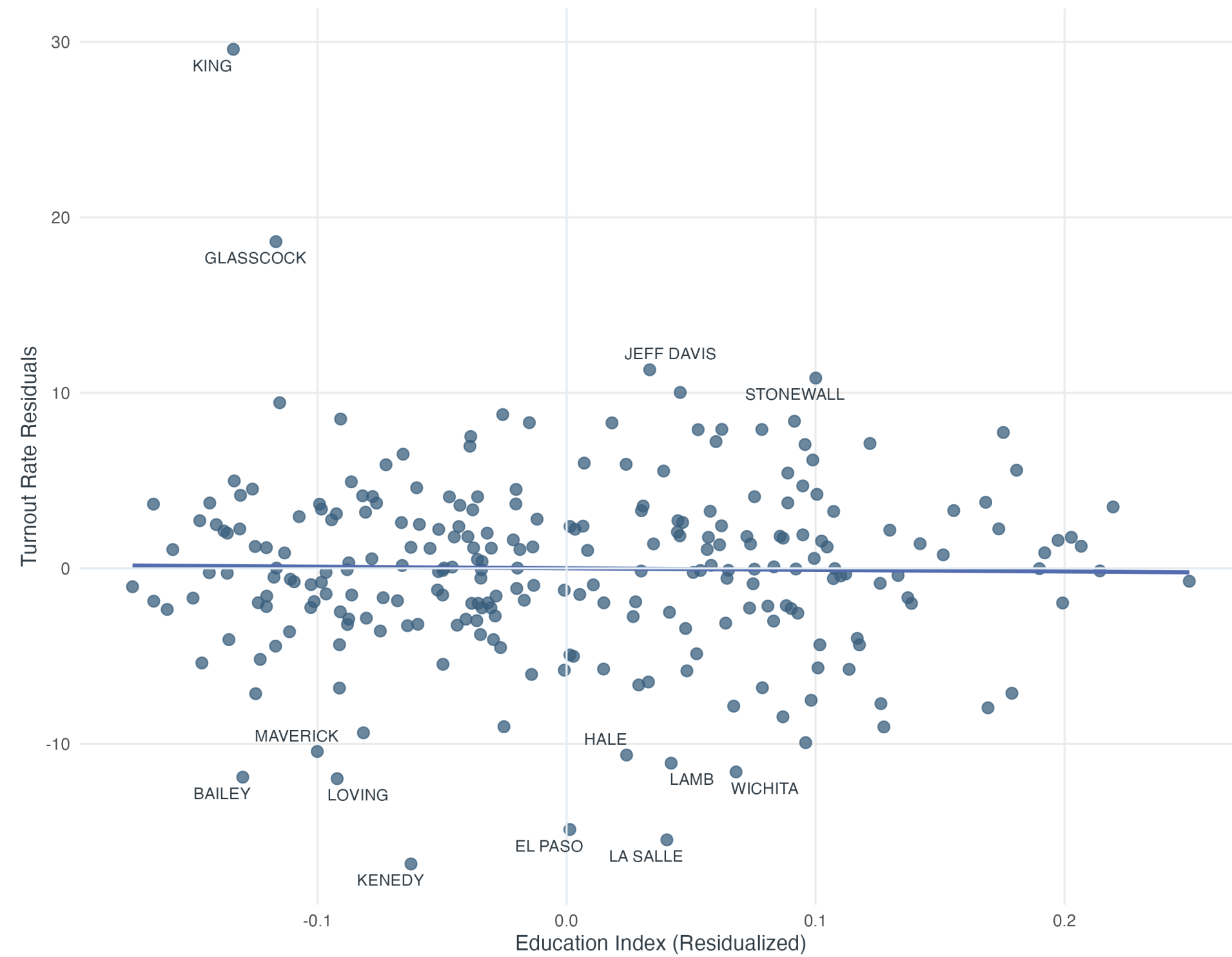


Turnout vs Health

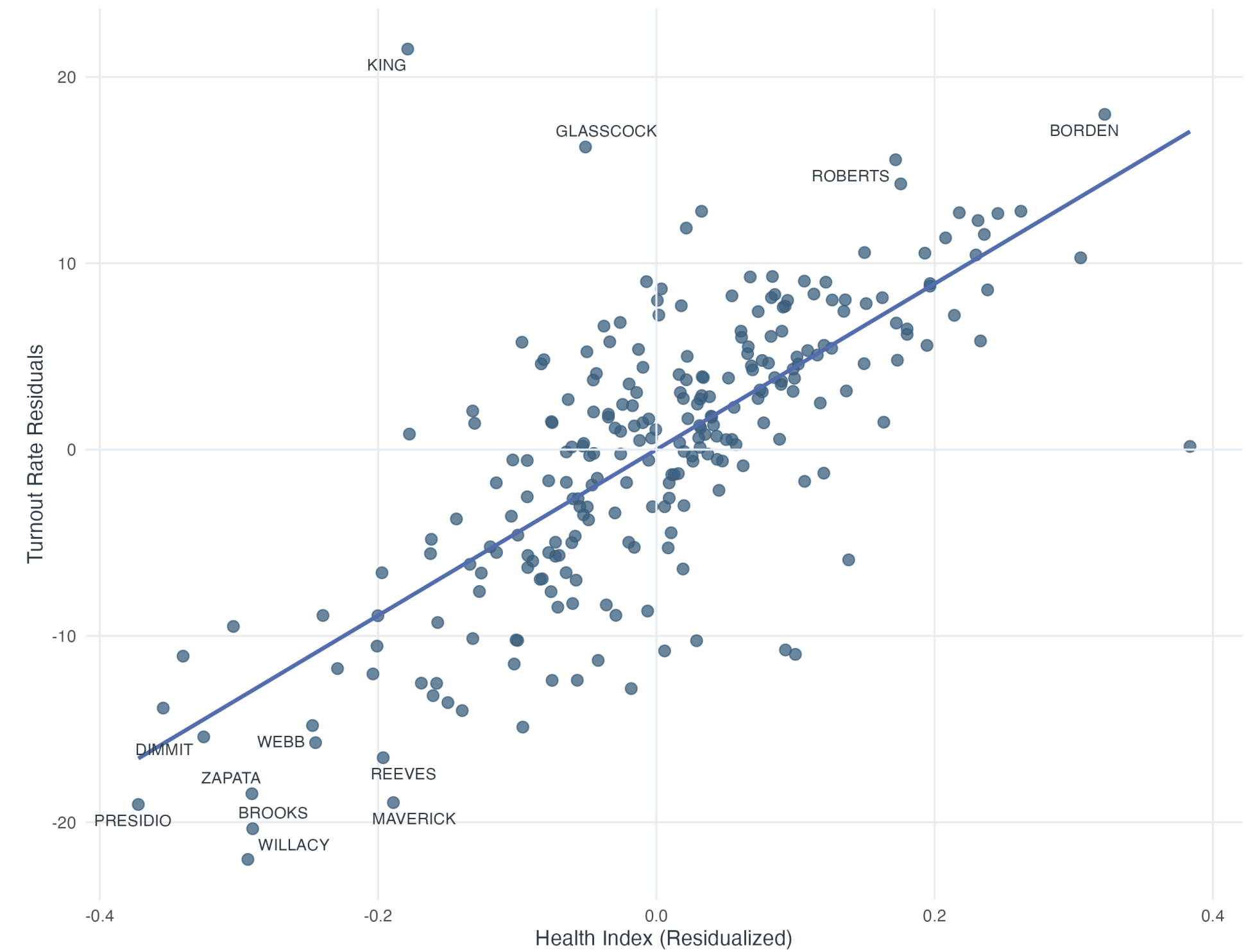


Turnout vs Education

Partial Regression



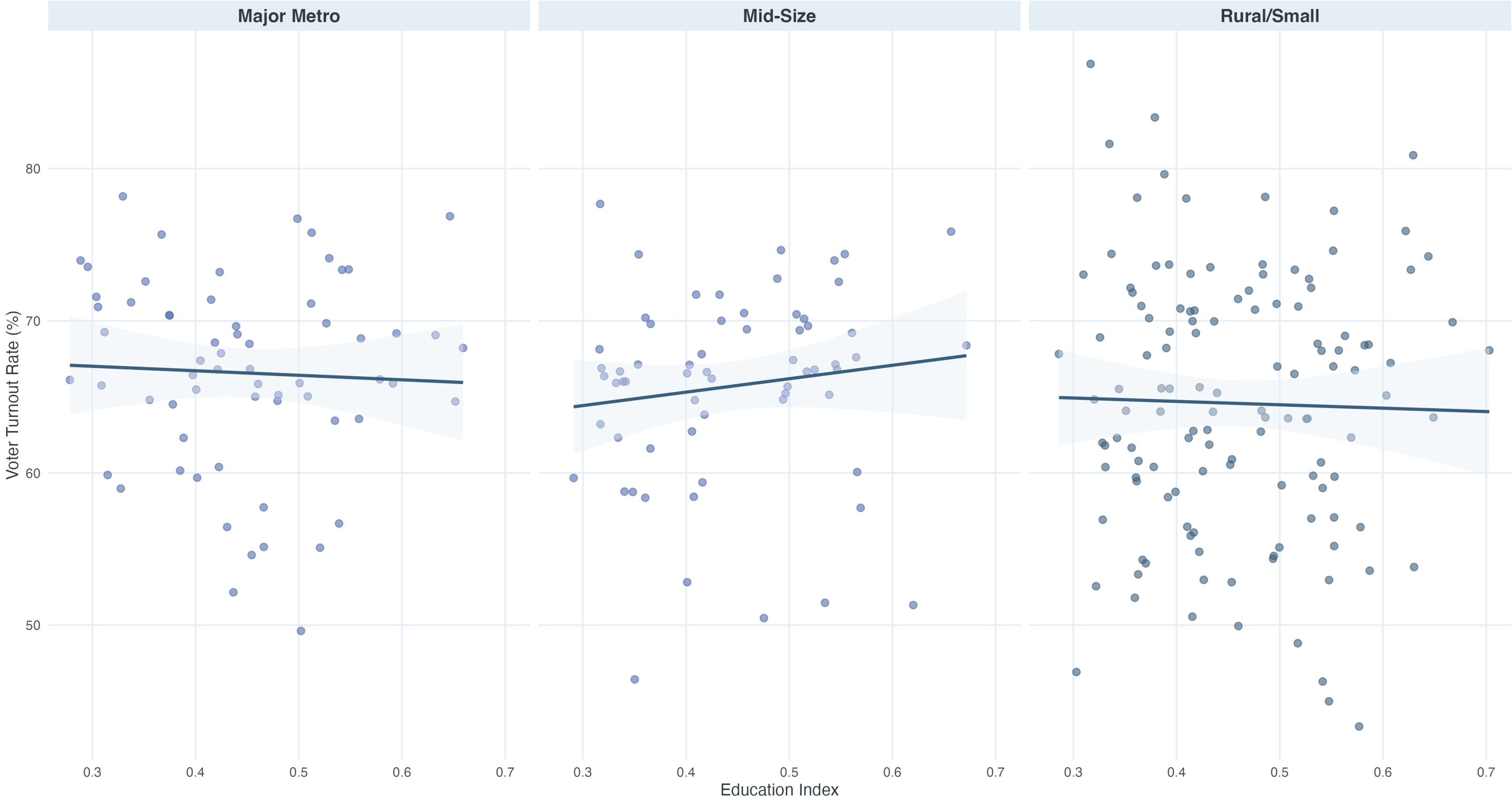
Education Controlling for Health



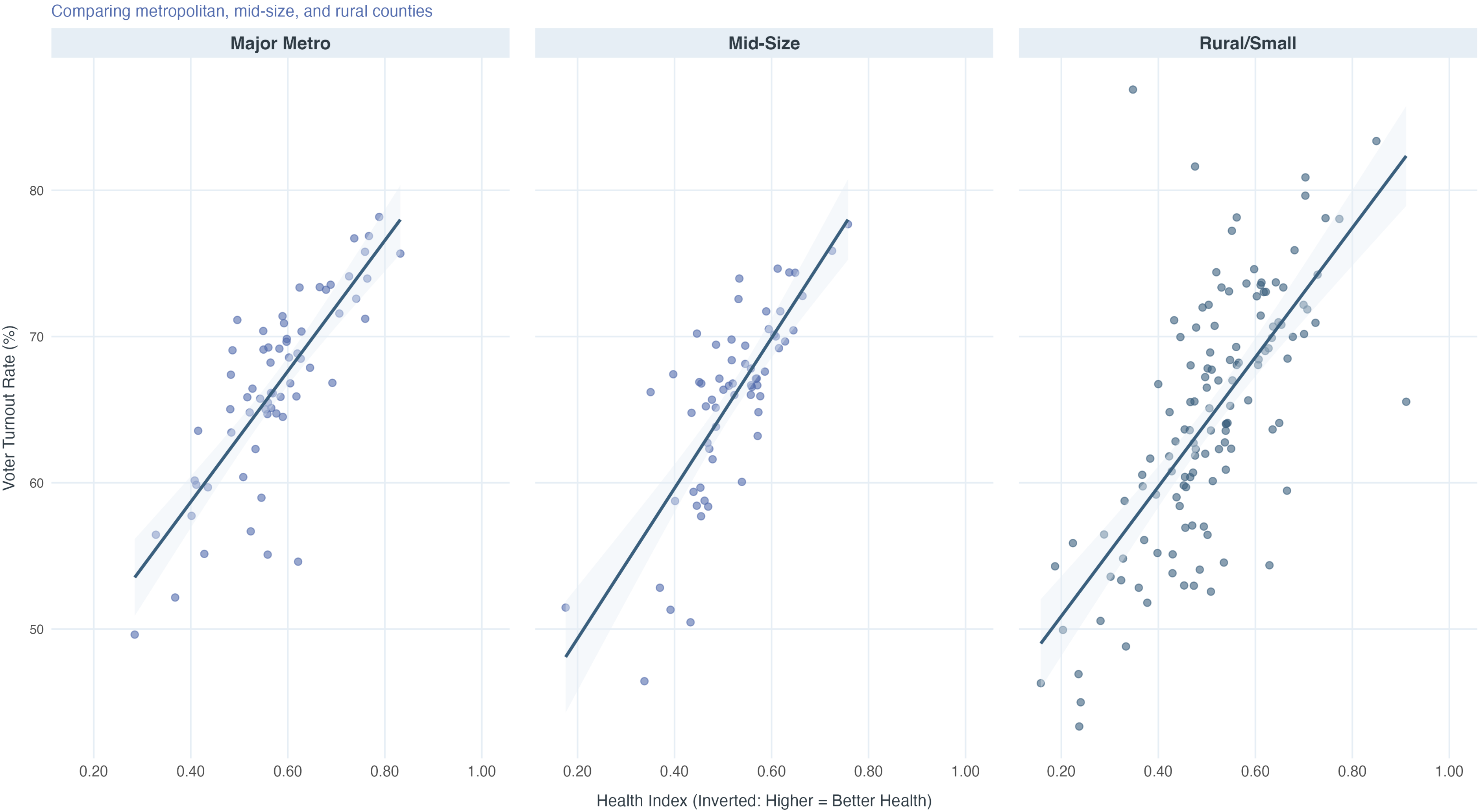
Health Controlling for Education

Counties / Education

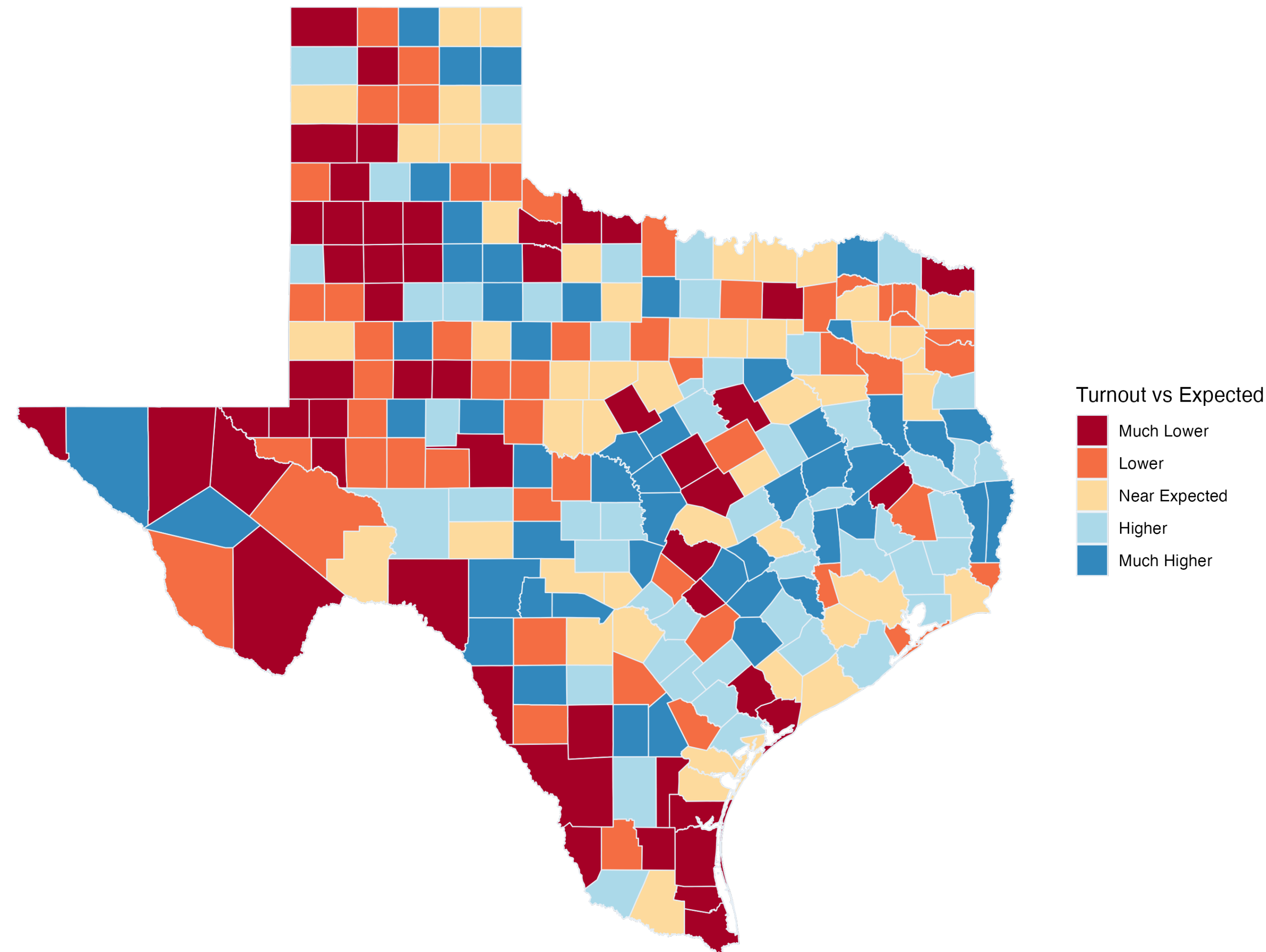
Comparing metropolitan, mid-size, and rural counties



Counties / Health

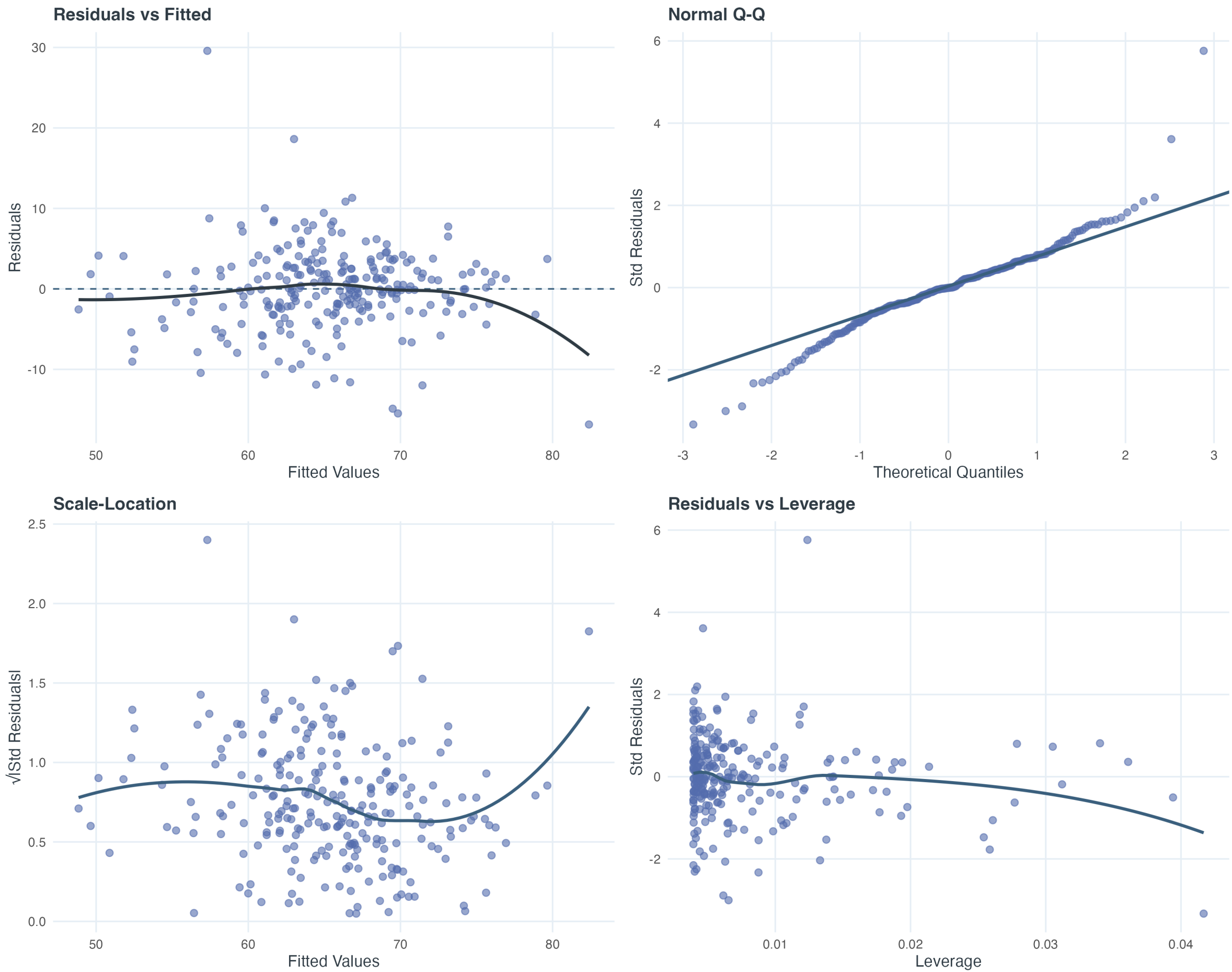


Regression Residuals

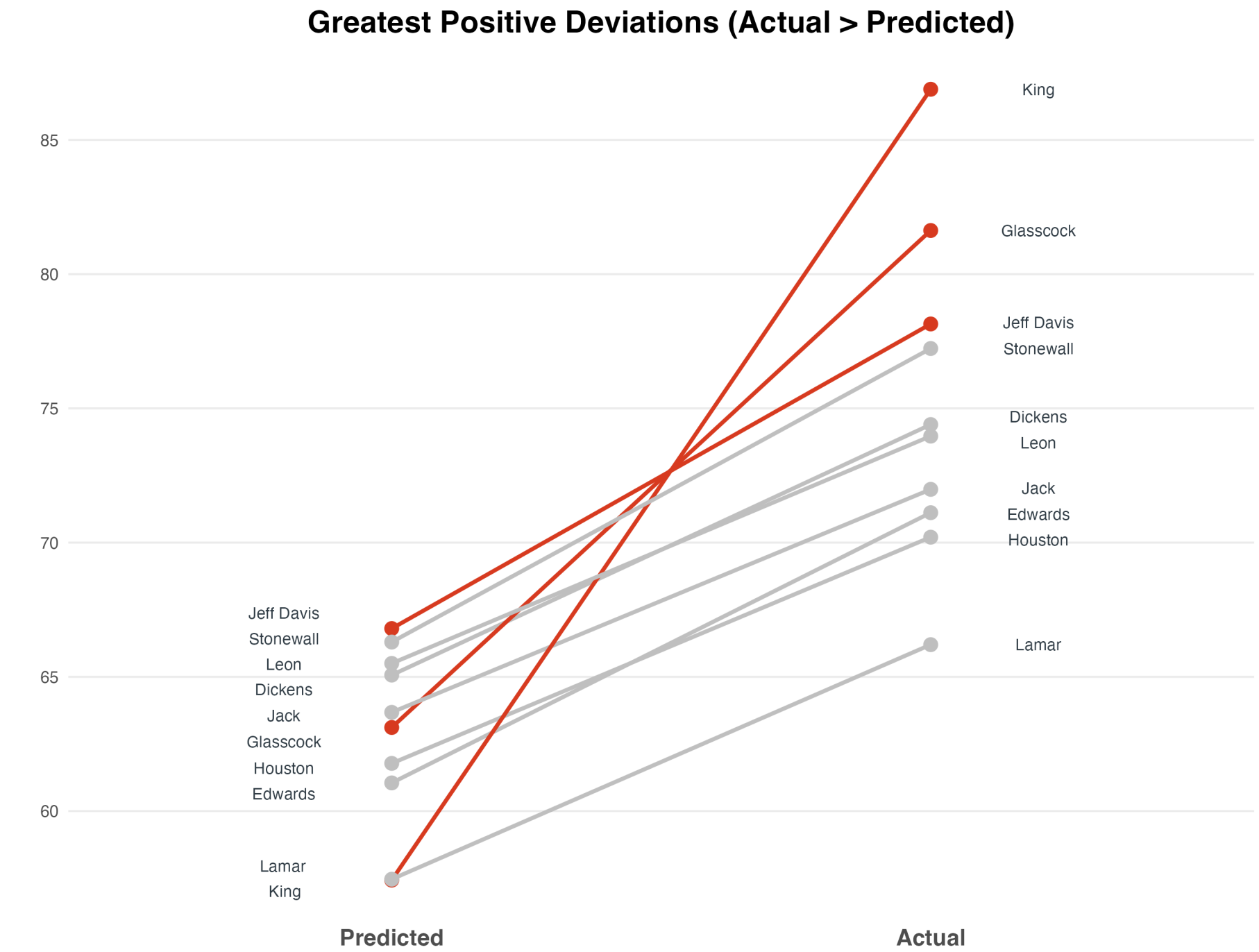
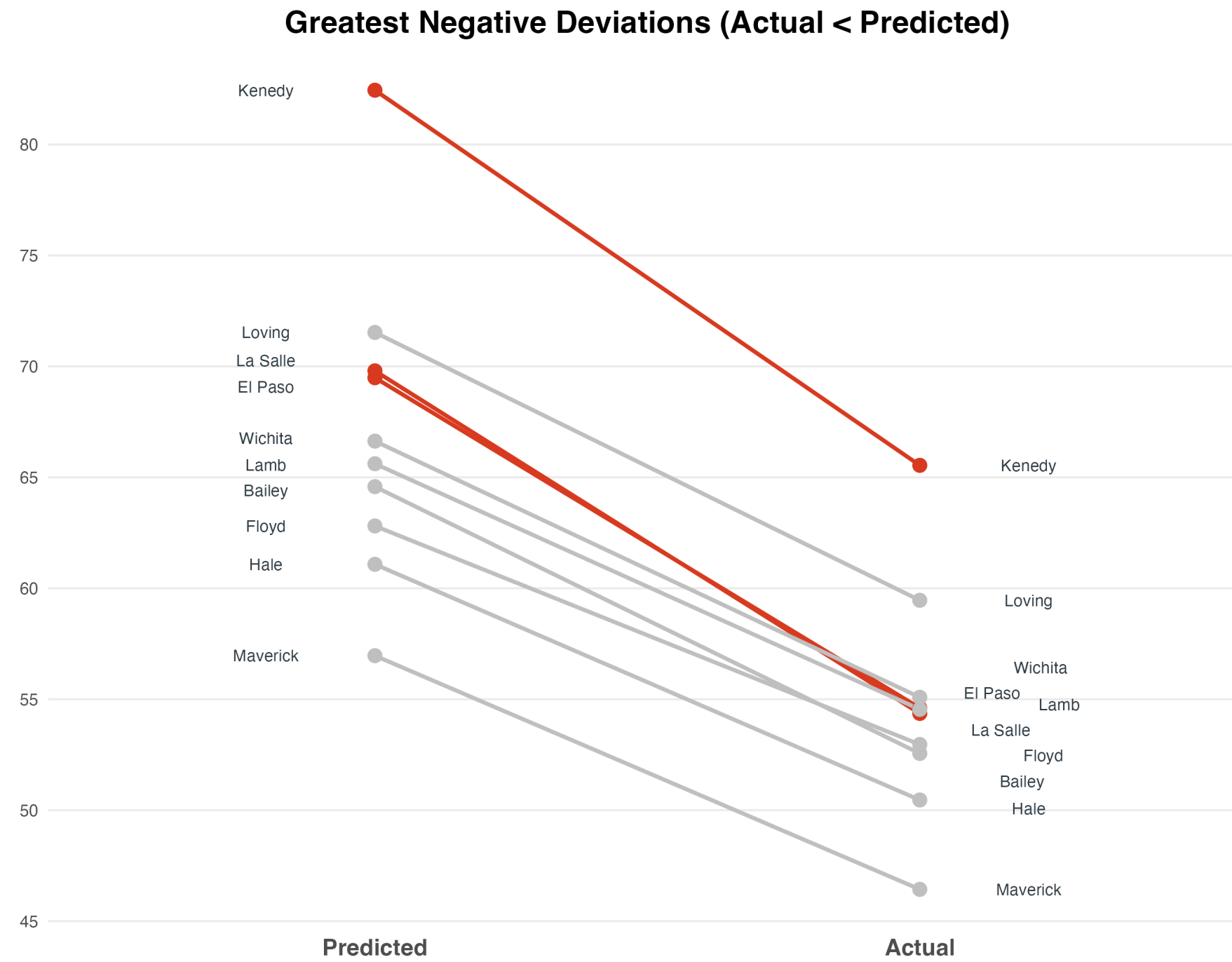


Model Diagnostics

Adjusted R² = 0.531 | Education excluded due to null relationship

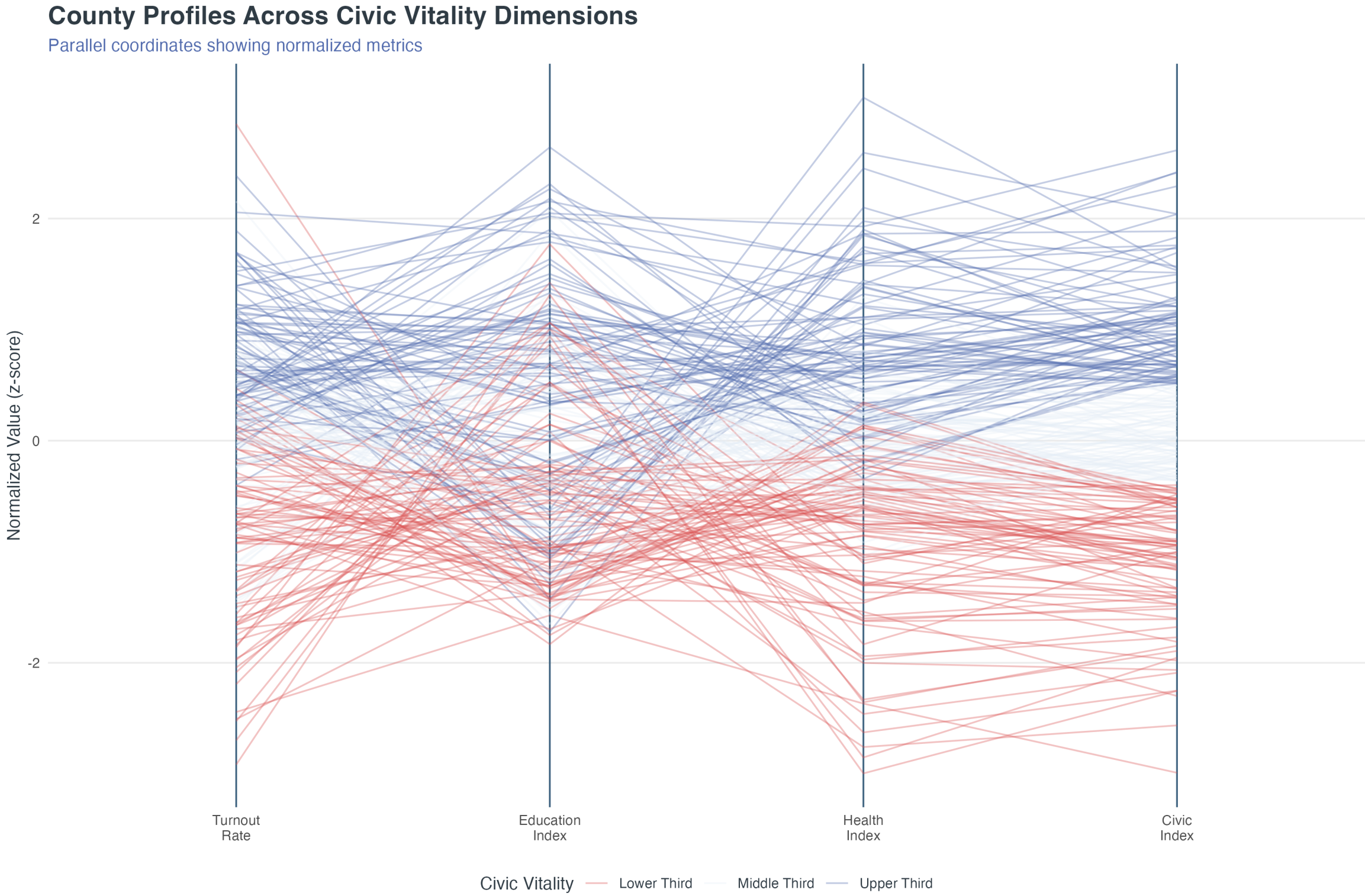


Model Deviations



- South Texas underperforms
- Panhandle overperforms
- Local or unaccounted factors are important

Model Deviations



Conclusions

- Health is the lever for civic participation
 - County health outcomes are the strongest structural predictor of voter turnout
 - This relationship holds across metro, mid-size, and rural counties
- Education's role is indirect (at best)
 - The effect of education documented in the literature does not seem to translate to county-level patterns
 - After controlling for health, education adds no explanatory power
- Geography shows systematic inequality
 - Civic gardens cluster around major metros with likely stronger health infrastructure
 - Civic deserts persist in South Texas, West Texas, and the Rio Grande Valley
 - Local political culture may create meaningful deviations from predictions
- Policy Implications:
 - Invest in community health infrastructure as a pathway to democratic participation
 - Target health disparities in underperforming regions (South/West Texas)
 - Don't assume education alone will boost turnout since health matters more

Healthy Communities Vote More